Tutorial 10 (Force Analysis)

(***Analytical method is to be used by default; the descriptions of the problems in the book are refined***)

<u>***Note**: You do not need to manually solve the equations in Problems 11-5 and 11-3. You are encouraged to use a software (e.g. MATLAB) to get numerical solutions.*</u>

- **Problem 11-5**

  Table P11-3 shows kinematic and geometric data for several pin-jointed fourbar linkages of the type and orientation shown in **Figure P11-2**. All have $\theta_1 = 0$. The point locations are defined as described in the text.

  For ***Row a*** in all the parts (Part 1, Part 2. Part 3 and Part 4) in the table assigned, use the matrix method of **Section 11.4** to derive the matrix equation to solve for forces and torques at the position shown.

TABLE P11-3 Data for Problems 11-5 and 11-7 (See Figure P11-2 for Nomenclature)

Part 1 Lengths in inches, angles in degrees, angular acceleration in rad/sec^2

Row	link 2	link 3	link 4	link 1	θ_2	θ_3	θ_4	α_2	α_3	α_4
a	4	12	8	15	45	24.97	99.30	20	75.29	244.43
b	3	10	12	6	30	90.15	106.60	– 5	140.96	161.75
c	5	15	14	2	260	128.70	151.03	15	78.78	53.37
d	6	19	16	10	– 75	91.82	124.44	– 10	– 214.84	– 251.82
e	2	8	7	9	135	34.02	122.71	25	71.54	– 14.19
f	17	35	23	4	120	348.08	19.01	– 20	– 101.63	– 150.86
g	7	25	10	19	100	4.42	61.90	– 15	– 17.38	–168.99

Part 2 Angular velocity in rad/sec, mass in blobs, moment of Inertia in blob-in^2, torque in lb-in

Row	ω_2	ω_3	ω_4	m_2	m_3	m_4	I_2	I_3	I_4	T_3	T_4
a	20	– 5.62	3.56	0.002	0.02	0.10	0.10	0.20	0.50	– 15	25
b	10	– 10.31	– 7.66	0.050	0.10	0.20	0.20	0.40	0.40	12	0
c	20	16.60	14.13	0.010	0.02	0.05	0.05	0.10	0.13	– 10	20
d	20	3.90	– 3.17	0.006	0.15	0.07	0.12	0.30	0.15	0	30
e	20	1.06	5.61	0.001	0.04	0.09	0.30	0.80	0.30	25	40
f	20	18.55	21.40	0.150	0.30	0.25	0.24	0.60	0.92	0	– 25
g	20	4.10	16.53	0.080	0.20	0.12	0.45	0.90	0.54	0	0

Part 3 Lengths in inches, angles in degrees, linear accelerations in in/sec^2

Row	R_{g2} mag	R_{g2} ang	R_{g3} mag	R_{g3} ang	R_{g4} mag	R_{g4} ang	a_{g2} mag	a_{g2} ang	a_{g3} mag	a_{g3} ang
a	2	0	5	0	4	30	801.00	222.14	1691.49	208.24
b	1	20	4	– 30	6	40	100.12	232.86	985.27	194.75
c	3	– 40	9	50	7	0	1200.84	37.85	3120.71	22.45
d	3	120	12	60	6	– 30	1200.87	226.43	4543.06	81.15
e	0.5	30	3	75	2	– 40	200.39	341.42	749.97	295.98
f	6	45	15	135	10	25	2403.00	347.86	12 064.20	310.22
g	4	– 45	10	225	4	45	1601.12	237.15	2562.10	–77.22

Part 4 Linear accelerations in in/sec^2, forces in lb, lengths in inches, angles in degrees

Row	a_{g4} mag	a_{g4} ang	F_{P3} mag	δF_{P3} ang	R_{P3} mag	δR_{P3} ang	F_{P4} mag	δF_{P4} ang	R_{P4} mag	δR_{P4} ang
a	979.02	222.27	0	0	0	0	40	– 30	8	0
b	1032.32	256.52	4	30	10	45	15	– 55	12	0
c	1446.58	316.06	0	0	0	0	75	45	14	0
d	1510.34	2.15	2	45	15	180	20	270	16	0
e	69.07	286.97	9	0	6	– 60	16	60	7	0
f	4820.72	242.25	0	0	0	0	23	0	23	0
g	1284.55	–41.35	12	– 60	9	120	32	20	10	0

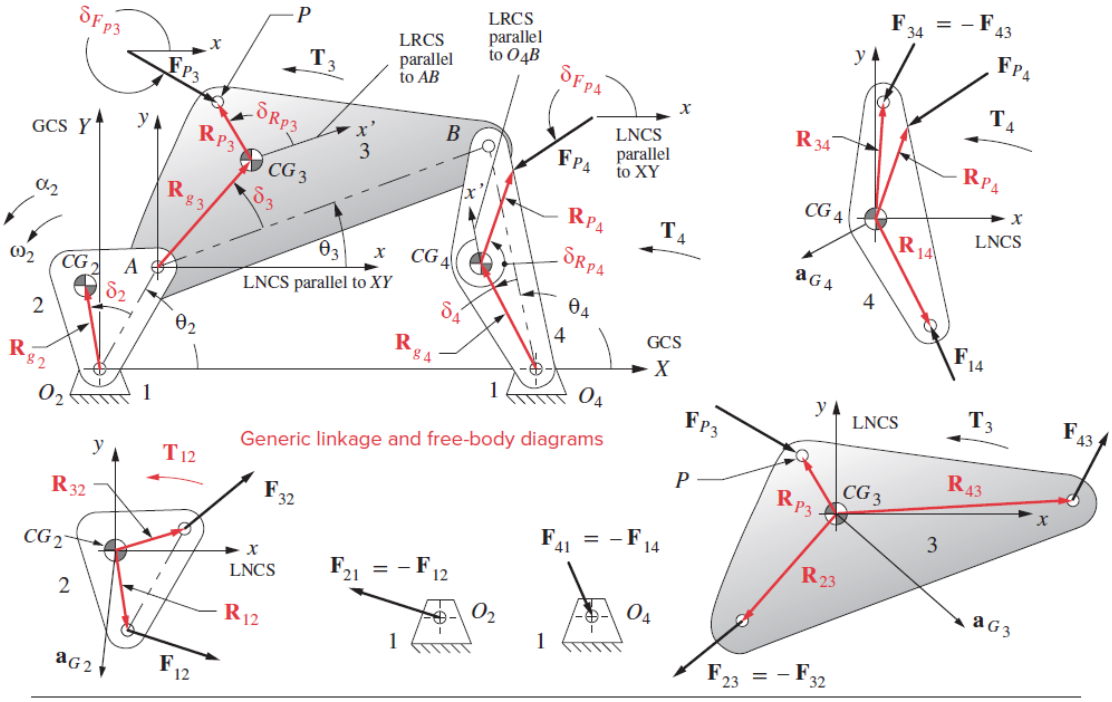


FIGURE P11-2

- ## Problem 11-6

  Repeat **Problem 11-5** using the method of virtual work to derive the equations to solve for the input torque on Link 2. Additional data for ***Row a*** are given in **Table P11-4**.

TABLE P11-4 Data for Problem 11-6

Row	Vg_2mag	Vg_2ang	Vg_3mag	Vg_3ang	Vg_4mag	Vg_4ang	$V_{P_3}mag$	$V_{P_3}ang$	$V_{P_4}mag$	$V_{P_4}ang$
a	40.00	135.00	54.44	145.19	14.23	219.30	54.44	145.19	41.39	–160.80
b	10.00	140.00	21.46	14.74	45.94	56.60	122.10	40.04	130.51	29.68
c	60.00	–50.00	191.94	299.70	98.91	241.03	191.94	–60.30	296.73	–118.97
d	60.00	135.00	94.36	353.80	19.03	4.44	152.51	–3.13	67.86	26.38
e	10.00	255.00	42.89	223.13	11.22	172.71	37.01	–140.37	48.41	–155.86
f	120.00	255.00	618.05	211.39	213.98	134.01	618.03	–148.61	692.08	116.52
g	80.00	145.00	118.29	205.52	66.10	196.90	154.85	–152.36	217.15	164.33

- ## Problem 11-3

  Table P11-1 shows kinematic and geometric data for several crank-slider linkages of the type and orientation shown in **Figure P11-1**. The point locations are defined as described in the text.

  For ***Row a*** in all the parts (Part 1, Part 2. Part 3 and Part 4) in the table assigned, use the matrix method of **Section 11.5** to derive the matrix equation to solve for forces and torques at the position shown.

TABLE P11-1 Data for Problem 11-3 (See Figure P11-1 for Nomenclature)

Part 1 — Lengths in inches, angles in degrees, mass in blobs, angular velocity in rad/sec

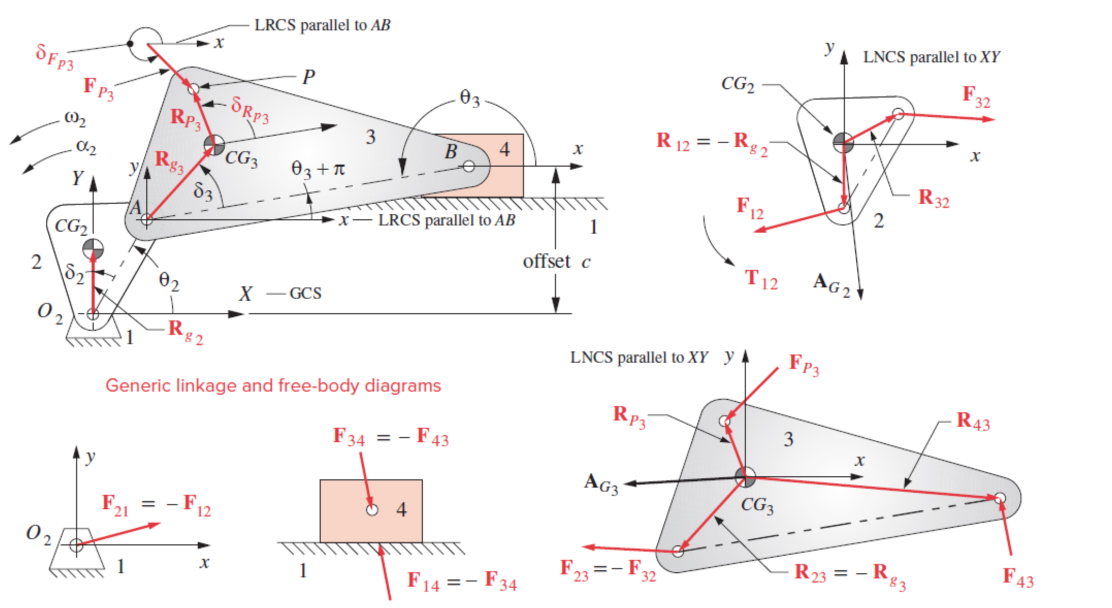
Row	link 2	link 3	offset	θ_2	ω_2	α_2	m_2	m_3	m_4
a	4	12	0	45	10	20	0.002	0.020	0.060
b	3	10	1	30	15	–5	0.050	0.100	0.200
c	5	15	–1	260	20	15	0.010	0.020	0.030
d	6	20	1	– 75	–10	– 10	0.006	0.150	0.050
e	2	8	0	135	25	25	0.001	0.004	0.014
f	10	35	2	120	5	– 20	0.150	0.300	0.050
g	7	25	–2	– 45	30	– 15	0.080	0.200	0.100

Part 2 — Angular acceleration in rad/sec^2, moments of Inertia in blob-in^2, torque in lb-in

Row	I_2	I_3	Rg_2 mag	δ_2 ang	Rg_3 mag	δ_3 ang	F_{P3} mag	δF_{P3} ang	R_{P3} mag	δR_P ang	T_3
a	0.10	0.2	2	0	5	0	0	0	0	0	20
b	0.20	0.4	1	20	4	– 30	10	45	4	30	– 35
c	0.05	0.1	3	– 40	9	50	32	270	0	0	– 65
d	0.12	0.3	3	120	12	60	15	180	2	60	– 12
e	0.30	0.8	0.5	30	3	75	6	– 60	2	75	40
f	0.24	0.6	6	45	15	135	25	270	0	0	– 75
g	0.45	0.9	4	– 45	10	225	9	120	5	45	– 90

Part 3 — Forces in lb, linear accelerations in in/sec^2

Row	θ_3	α_3	a_{g2} mag	a_{g2} ang	a_{g3} mag	a_{g3} ang	a_{g4} mag	a_{g4} ang
a	166.40	– 2.40	203.96	213.69	371.08	200.84	357.17	180
b	177.13	34.33	225.06	231.27	589.43	200.05	711.97	180
c	195.17	–134.76	1200.84	37.85	2088.04	43.43	929.12	0
d	199.86	– 29.74	301.50	230.71	511.74	74.52	23.97	180
e	169.82	113.12	312.75	–17.29	976.79	–58.13	849.76	0
f	169.03	3.29	192.09	23.66	302.50	–29.93	301.92	0
g	186.78	– 172.20	3600.50	90.95	8052.35	134.66	4909.27	180



Generic linkage and free-body diagrams

FIGURE P11-1

- ## Problem 11-4

  Repeat **Problem 11-3** using the method of virtual work to derive the equations to solve for the input torque on Link 2. Additional data for ***Row a*** are given in **Table P11-2**.

TABLE P11-2 Data for Problem 11-4

See also Table P11-1. Unit system is the same as in that table.

Row	ω_3	$Vg_2 mag$	$Vg_2 ang$	$Vg_3 mag$	$Vg_3 ang$	$Vg_4 mag$	$Vg_4 ang$	$V_{P_3} mag$	$V_{P_3} ang$
a	– 2.43	20.0	135	35.24	152.09	35.14	180	35.24	152.09
b	– 3.90	15.0	140	40.35	140.14	24.45	180	26.69	153.35
c	1.20	60.0	310	89.61	–8.23	93.77	0	89.61	– 8.23
d	0.83	30.0	315	69.10	191.15	63.57	180	70.63	191.01
e	4.49	12.5	255	56.02	211.93	29.01	180	61.36	204.87
f	0.73	30.0	255	60.89	210.72	38.46	180	60.89	210.72
g	–5.98	120.0	0	211.46	61.31	166.14	0	208.60	53.19